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Design and Implementation of a Two-Stage Single-Phase Pure Sine Wave Inverter using STM32G4

This project is submitted as partial fulfillment of the requirement for the Power Electronics and Drives studio for Spring 2025-26.

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DECLARATION

We hereby declare that the report entitled “Two-Stage Single-Phase Pure Sine Wave Inverter using STM32G4” is a bonafide record of the work carried out by us in partial fulfillment of the requirements for the completion of the 6th semester Power Electronics and Drives Studio of Bachelor of Technology in Electrical and Electronics Engineering.

We further declare that this work has been carried out by us under the guidance of our supervisor and has not been submitted elsewhere for the completion of any other degree, diploma, or fellowship. All sources of information used in this report have been duly acknowledged.

We also affirm that this project is an original work and has been completed by us to the best of our knowledge and belief.

Signature of the Guide

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to everyone who supported and guided us throughout the completion of our project titled “Design and Implementation of a Two-Stage Single-Phase Pure Sine Wave Inverter using STM32G4.”

We are highly thankful to Navrachana University for providing us with the opportunity, facilities, and resources required to successfully carry out this project work. The academic environment and technical support offered by the university greatly contributed to the successful completion of this project.

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Finally, we would like to thank all those who directly or indirectly contributed to the completion of this project. Without their support and encouragement, this work would not have been possible.

ABSTRACT

This project presents the design and implementation of a Two-Stage Single-Phase Pure Sine Wave Inverter using the STM32G4 microcontroller for efficient DC to AC power conversion. The system is designed to generate a stable and low-distortion single-phase AC output suitable for domestic and industrial low-power applications. The inverter employs a two-stage conversion topology consisting of a rectification stage and an inverter stage to achieve improved voltage regulation, waveform quality, and operational reliability.

In the first stage, the AC input supply is converted into DC using an uncontrolled diode bridge rectifier. The rectified output is filtered using a capacitor bank to obtain a stable DC link voltage. In the second stage, a full-bridge inverter topology is implemented using MOSFET switches controlled by the STM32G4 microcontroller. The controller generates Sinusoidal Pulse Width Modulation (SPWM) signals to ensure accurate switching operation and to produce a sinusoidal output waveform with reduced harmonic distortion.

An LC low-pass filter is incorporated at the output stage to eliminate high-frequency switching harmonics and improve the quality of the generated sine wave. The proposed system is capable of delivering a 230 V RMS, 50 Hz single-phase AC output with efficient power conversion and reliable performance. The use of the STM32G4 microcontroller enables precise PWM generation, fast processing capability, flexible control implementation, and enhanced system stability.

The developed inverter system demonstrates advantages such as compact design, reduced Total Harmonic Distortion (THD), improved efficiency, electrical isolation, and scalability for future enhancements. This project highlights the practical implementation of modern power electronic conversion techniques and embedded control systems for pure sine wave inverter applications.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AC	Alternating Current
ADC	Analog to Digital Converter
DC	Direct Current
EMI	Electromagnetic Interference
ESR	Equivalent Series Resistance
IC	Integrated Circuit
IoT	Internet of Things
LC Filter	Inductor-Capacitor Filter
MOSFET	Metal Oxide Semiconductor Field Effect Transistor
MCU	Microcontroller Unit
PCB	Printed Circuit Board
PI Controller	Proportional Integral Controller
PSIM	Power Simulation Software
PWM	Pulse Width Modulation
RC Circuit	Resistor Capacitor Circuit
RCD Clamp	Resistor Capacitor Diode Clamp
RMS	Root Mean Square
SPWM	Sinusoidal Pulse Width Modulation

THD	Total Harmonic Distortion
UPS	Uninterruptible Power Supply
VDC	Direct Current Voltage
VAC	Alternating Current Voltage
Hz	Hertz
kHz	Kilohertz
STM32G4	STMicroelectronics 32-bit Microcontroller Series
GPIO	General Purpose Input Output
ADC	Analog to Digital Converter
ESR	Equivalent Series Resistance
H-Bridge	Full Bridge Inverter Configuration
CPU	Central Processing Unit
USB	Universal Serial Bus

LIST OF APPENDIX

CHAPTER 1

INTRODUCTION

1.1 Background

The increasing demand for reliable and high-quality electrical power has led to the rapid development of power electronic conversion systems. In modern electrical and industrial applications, inverters play a vital role in converting Direct Current (DC) power into Alternating Current (AC) power suitable for domestic, commercial, and industrial loads [1]. Conventional square wave and modified sine wave inverters introduce significant harmonic distortion, reduced efficiency, and poor compatibility with sensitive electronic equipment. Therefore, Pure Sine Wave Inverters have become increasingly important due to their ability to provide clean, stable, and low-distortion AC output similar to utility power systems [2].

A two-stage inverter topology is widely preferred in modern power conversion systems because of its improved voltage regulation, flexibility, efficiency, and control capability [3]. In a two-stage architecture, the first stage performs AC to DC conversion and establishes a stable DC bus, while the second stage converts the DC voltage into a controlled sinusoidal AC output. This approach improves waveform quality, simplifies control implementation, and enhances system stability under varying load conditions [4].

The proposed project focuses on the design and implementation of a Two-Stage Single-Phase Pure Sine Wave Inverter using the STM32G4 microcontroller. The system utilizes a three-phase AC input supply which is fed through a variac for adjustable voltage control. In the first stage, a three-phase full-bridge rectifier converts the AC input into a high-voltage DC output of approximately 400 V DC. The rectified voltage is filtered using a capacitor bank to create a stable DC link for the inverter stage [1].

In the second stage, a full-bridge inverter topology operating at a switching frequency of 20 kHz is implemented using MOSFET switches. The inverter is controlled using Sinusoidal Pulse Width Modulation (SPWM) signals generated by the STM32G4 microcontroller [3]. The high-frequency PWM switching technique allows the inverter to produce an output waveform that closely resembles a pure sine wave. An LC low-pass filter is incorporated at the output stage to eliminate high-frequency harmonics and improve waveform quality [8].

The STM32G4 microcontroller is selected due to its advanced timer peripherals, high-speed processing capability, accurate PWM generation, and efficient real-time control features [5]. The controller also performs voltage and current sensing, PI-based control implementation, fault protection, and system monitoring to ensure safe and reliable operation.

The developed inverter system is designed to generate a stable 230 V, 50 Hz single-phase AC output, making it suitable for residential backup systems, laboratory applications, renewable energy systems, and low-power industrial loads. The project demonstrates the practical implementation of modern embedded control systems and power electronics techniques for efficient energy conversion [2], [4].

1.2 Project Objective

The primary objective of this project is to design and implement a Two-Stage Single-Phase Pure Sine Wave Inverter using the STM32G4 microcontroller capable of producing a stable and low-distortion AC output from a DC source.

The specific objectives of the project are as follows:

1. To design a two-stage power conversion system consisting of a rectification stage and an inverter stage.
2. To convert a three-phase AC supply into a stable high-voltage DC bus using a three-phase full-bridge rectifier and capacitor filtering network [1].
3. To generate a pure sinusoidal AC output of 230 V RMS at 50 Hz using a full-bridge inverter topology.
4. To implement Sinusoidal Pulse Width Modulation (SPWM) techniques using the STM32G4 microcontroller for accurate switching control of MOSFET devices [3], [5].
5. To reduce harmonic distortion in the output waveform using an LC low-pass filter [8].
6. To implement current sensing, voltage sensing, protection mechanisms, and PI control algorithms for improved system reliability and stability.
7. To achieve efficient power conversion with reduced switching losses and improved waveform quality [4].

8. To develop a compact, modular, and scalable inverter system suitable for practical applications and future enhancements.

1.3 Project Scope

The scope of this project includes the complete design, development, testing, and implementation of a two-stage single-phase pure sine wave inverter system using embedded control techniques and power electronic converters.

The project covers the following areas:

1. Design and implementation of the power stage consisting of a three-phase rectifier and full-bridge inverter topology [1].
2. Development of a stable 400 V DC bus using rectification and capacitor filtering techniques.
3. Generation of high-frequency SPWM signals using the STM32G4 microcontroller [5].
4. Design of gate driver circuits for MOSFET switching applications.
5. Implementation of an LC output filter for harmonic reduction and waveform smoothing [8].
6. Integration of voltage and current sensing circuits for feedback and protection.
7. Implementation of PI control algorithms for output regulation and system stability.
8. Testing and analysis of inverter performance under different operating conditions and load variations.
9. Measurement and evaluation of output voltage quality, switching performance, efficiency, and harmonic distortion.

The proposed inverter system can be further extended for applications such as renewable energy integration, solar inverter systems, UPS systems, battery energy storage systems, and industrial motor drive applications [6]. Future improvements may include closed-loop digital control, advanced modulation techniques, higher power ratings, IoT-based monitoring, and improved efficiency optimization methods.

CHAPTER 2

LITERATURE REVIEW

2.1 Pure Sine Wave Inverter

An inverter is a power electronic device used to convert Direct Current (DC) into Alternating Current (AC). Inverters are widely used in backup power systems, renewable energy systems, industrial drives, and domestic applications. According to Mohan et al. [1], modern inverter systems mainly focus on improving efficiency, reducing switching losses, and minimizing harmonic distortion in the output waveform.

There are mainly three types of inverter output waveforms: square wave, modified sine wave, and pure sine wave. Among these, pure sine wave inverters provide the best output quality because the generated waveform closely matches the utility AC supply [2]. Pure sine wave inverters are preferred for sensitive electronic devices, medical equipment, communication systems, and industrial loads due to their low Total Harmonic Distortion (THD) and stable operation.

The proposed project implements a pure sine wave inverter using a two-stage conversion system. The first stage converts AC input into DC using a rectifier circuit, while the second stage converts the DC link voltage into sinusoidal AC output using SPWM switching techniques.

2.2 Two-Stage Inverter Topology

The two-stage inverter topology is widely used in modern power conversion systems because of its improved flexibility, voltage regulation capability, and waveform quality [4]. In this topology, the first stage consists of an AC-DC converter or rectifier that generates a stable DC bus voltage. The second stage consists of a DC-AC inverter that produces the required AC output waveform.

In the proposed system, a three-phase full-bridge rectifier is used to convert the three-phase AC supply into approximately 400 V DC. A capacitor bank is connected across the DC link to reduce voltage ripple and maintain a stable DC bus voltage. The inverter stage uses a full-bridge MOSFET configuration operating at high switching frequency to generate AC output voltage.

The two-stage structure provides better isolation between rectification and inversion processes, improves output stability, and allows independent control of both stages. This topology is commonly used in UPS systems, renewable energy systems, and high-performance inverter applications [1], [4].

2.3 Sinusoidal Pulse Width Modulation (SPWM)

Pulse Width Modulation (PWM) is one of the most important switching techniques used in inverter systems. PWM techniques are used to control switching devices and generate output voltages with reduced harmonic distortion [3]. Among different PWM techniques, Sinusoidal Pulse Width Modulation (SPWM) is widely used because of its simple implementation and high-quality output waveform generation.

Holmes and Lipo [3] explained that SPWM is generated by comparing a high-frequency triangular carrier signal with a sinusoidal reference signal. The comparison produces PWM pulses that control the switching of MOSFETs in the inverter circuit. By varying the modulation index and switching frequency, the output voltage magnitude and waveform quality can be controlled.

SPWM significantly reduces lower-order harmonics and improves inverter efficiency. High-frequency switching also reduces the size of output filters required for harmonic reduction [8]. In this project, the STM32G4 microcontroller generates 20 kHz SPWM signals to control the inverter switching operation.

2.4 STM32G4 Microcontroller

Microcontrollers play an important role in modern inverter systems due to their ability to provide accurate switching control, sensing, protection, and feedback implementation. The STM32G4 series microcontroller developed by STMicroelectronics [5] is specifically designed for power electronic and motor control applications.

The STM32G4 controller provides advanced timer peripherals, high-speed Analog-to-Digital Converters (ADC), PWM generation units, dead-time insertion capability, and fault handling mechanisms. These features make the controller highly suitable for inverter applications requiring high-frequency switching and real-time control.

In the proposed project, the STM32G4 microcontroller is used for SPWM pulse generation, current and voltage sensing, PI control implementation, and protection handling. The controller ensures precise switching operation and stable output waveform generation.

2.5 MOSFET Switching Devices

Power semiconductor devices are the key components in inverter systems. MOSFETs are widely used in high-frequency inverter applications because of their fast switching speed, low switching losses, and simple gate drive requirements [2].

According to Ahmed [9], MOSFET-based inverter systems provide improved switching performance and higher efficiency in medium-power applications. MOSFETs are voltage-controlled devices and require comparatively lower gate drive power than conventional bipolar transistors.

In the proposed project, MOSFETs are used in a full-bridge inverter configuration for high-frequency switching operation. The switching signals generated by the STM32G4 controller control the MOSFET operation to generate SPWM output pulses.

2.6 LC Output Filter

The output waveform of an inverter contains high-frequency switching harmonics generated due to PWM operation. Therefore, filters are required to reduce harmonic content and improve waveform quality. LC low-pass filters are commonly used in pure sine wave inverter systems because of their ability to attenuate high-frequency harmonics while allowing the fundamental frequency component to pass [8].

Ang and Oliva [10] explained that proper filter design is important for achieving stable output voltage and minimizing distortion. The inductance and capacitance values must be selected carefully based on switching frequency, load conditions, and output voltage requirements.

In this project, an LC filter is connected at the inverter output stage to smooth the SPWM waveform and generate a clean 230 V, 50 Hz sinusoidal AC output.

2.7 Control and Protection System

Modern inverter systems require efficient control and protection mechanisms to ensure safe and reliable operation. Feedback control techniques such as PI controllers are widely used for maintaining stable output voltage under varying load conditions [6].

Krishnan [6] explained that voltage sensing and current sensing circuits are necessary for monitoring system performance and implementing closed-loop control. Protection systems are also required to prevent damage caused by overcurrent, short circuits, overheating, and abnormal operating conditions.

The proposed inverter system incorporates voltage sensing, current sensing, PI control algorithms, and fault handling techniques using the STM32G4 microcontroller. These features improve system stability, reliability, and operational safety.

2.8 Summary of Literature Review

Based on the literature survey, it is observed that modern pure sine wave inverter systems require efficient PWM techniques, high-frequency switching devices, advanced digital controllers, and effective filtering methods for achieving high-quality AC power conversion. Most researchers focused on improving inverter efficiency, reducing harmonic distortion, and implementing reliable embedded control systems.

The concepts studied from the reviewed literature were used in the development of the proposed project, which implements a two-stage pure sine wave inverter using a three-phase rectifier, STM32G4-based SPWM control, MOSFET switching, and LC filtering techniques.

CHAPTER 3

SYSTEM DESIGN APPROACH

3.1 Design Objective

The primary objective of the proposed system is to design and implement a Two-Stage Single-Phase Pure Sine Wave Inverter using the STM32G4 microcontroller capable of generating a stable and low-distortion AC output suitable for practical electrical applications. The system is designed to convert a three-phase AC input into a regulated DC voltage and further convert the DC voltage into a single-phase sinusoidal AC output using high-frequency switching techniques [1], [4].

The design aims to achieve efficient power conversion, reduced harmonic distortion, stable voltage regulation, and reliable switching operation. Another major objective is to implement Sinusoidal Pulse Width Modulation (SPWM) using the STM32G4 microcontroller for accurate control of MOSFET switching devices [3], [5]. The proposed system also focuses on implementing voltage sensing, current sensing, output filtering, and protection mechanisms to improve overall system stability and reliability [4], [6].

The inverter is designed to provide a 230 V RMS, 50 Hz pure sine wave output with improved waveform quality and reduced Total Harmonic Distortion (THD) [1], [2], [8].

3.2 Selection of Two-Stage Topology

The two-stage inverter topology was selected because of its improved flexibility, better voltage regulation capability, modular structure, and enhanced waveform quality compared to conventional single-stage inverter systems [4]. In a two-stage architecture, the power conversion process is divided into two separate stages: the rectification stage and the inversion stage.

In the first stage, a three-phase full-bridge rectifier converts the AC input supply into a high-voltage DC output. The rectified voltage is filtered using a capacitor bank to create a stable DC bus voltage. In the second stage, the DC voltage is converted into single-phase AC output using a full-bridge inverter operating with high-frequency SPWM switching [1], [3].

The two-stage topology provides several advantages such as improved output voltage stability, better harmonic reduction capability, independent control of rectifier and inverter stages, easier implementation of digital control techniques, improved scalability and modularity, and better suitability for high-frequency switching applications [4], [6]. Due to these advantages, the two-stage topology is widely used in UPS systems, renewable energy systems, and modern pure sine wave inverter applications [1], [2].

3.3 Block Diagram and System Flow

The overall system consists of three major sections: the power stage, inverter stage, and control stage. The system begins with a three-phase AC supply connected through a variac for adjustable input voltage control. The AC input is fed to a three-phase full-bridge rectifier which converts the AC voltage into approximately 400 V DC.

The rectified DC output is filtered using a capacitor bank to reduce voltage ripple and establish a stable DC link voltage. This DC bus is then supplied to the inverter stage consisting of a full-bridge MOSFET configuration operating with SPWM switching at 20 kHz.

The STM32G4 microcontroller generates SPWM pulses required for MOSFET switching control. The inverter output is passed through an LC low-pass filter to eliminate switching harmonics and generate a clean sinusoidal AC waveform. Voltage sensing and current sensing circuits continuously monitor the system parameters and provide feedback to the controller for stable operation and protection handling.

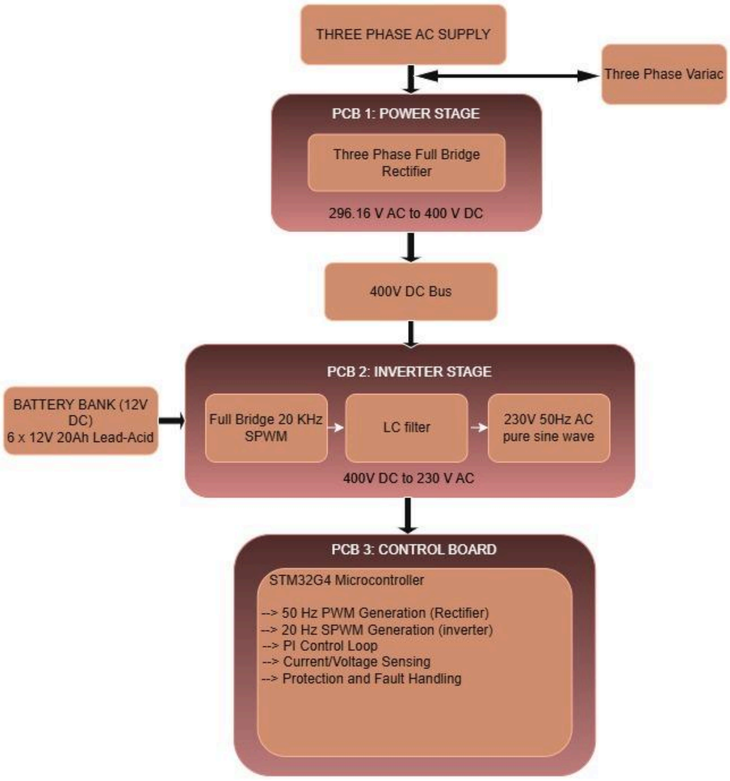


Fig 3.3.1 System Flow Chart

3.4 DC Link Design Concept

The DC link stage plays an important role in maintaining stable inverter operation and reducing voltage ripple between the rectifier and inverter stages. The DC link consists of a capacitor bank connected across the rectifier output to store energy and smooth the pulsating DC voltage generated by the rectifier [1], [4].

The capacitor bank helps in reducing ripple voltage, stabilizing DC bus voltage, improving inverter performance, providing temporary energy storage during switching operation, and reducing voltage fluctuations caused by load variations. Proper DC link design is essential for maintaining stable inverter operation and improving overall power conversion efficiency [2], [6].

The DC bus voltage for the proposed system is designed around 400 V DC. Proper capacitor selection is important because insufficient capacitance may increase ripple voltage and affect inverter output quality. High-voltage electrolytic capacitors with suitable ripple current ratings are used to ensure stable operation [1], [4].

The DC link design also improves transient response and ensures continuous power delivery to the inverter stage during high-frequency switching operation [3], [8].

3.5 Switching Strategy Selection (SPWM)

Sinusoidal Pulse Width Modulation (SPWM) was selected as the switching strategy for the proposed inverter system because of its simplicity, effectiveness, and ability to generate high-quality sinusoidal output waveforms with reduced harmonic distortion [3].

In SPWM, a high-frequency triangular carrier signal is compared with a low-frequency sinusoidal reference signal. The comparison generates PWM pulses that control the switching operation of the MOSFETs in the full-bridge inverter circuit. By varying the modulation index and switching frequency, the output voltage magnitude and waveform quality can be controlled [3], [8].

The major advantages of SPWM include reduced Total Harmonic Distortion (THD), better output voltage control, improved waveform quality, reduced filter size requirements, higher inverter efficiency, and simple implementation using digital controllers [2], [3].

The STM32G4 microcontroller generates 20 kHz SPWM signals with proper dead-time insertion to avoid shoot-through conditions in the inverter switches. The modulation index and switching frequency are selected carefully to achieve stable output voltage and efficient switching performance [5], [8].

3.6 Output Filtering Approach

The inverter output produced by SPWM switching contains high-frequency harmonics due to rapid switching operation of the MOSFETs. Therefore, an output filter is required to eliminate switching harmonics and obtain a smooth sinusoidal waveform [3], [8].

An LC low-pass filter is used in the proposed system because of its effective harmonic attenuation capability and simple design structure. The inductor opposes sudden current changes while the capacitor smooths voltage fluctuations, resulting in a cleaner output waveform [8], [10].

The LC filter is designed based on switching frequency, output frequency, load conditions, harmonic attenuation requirements, and voltage and current ratings.

Proper filter parameter selection is important for maintaining waveform quality and reducing Total Harmonic Distortion (THD) [3], [10].

The filter significantly reduces high-frequency switching components and improves the quality of the generated 230 V, 50 Hz AC output waveform [2], [8].

3.7 Design Specifications

The major design specifications of the proposed inverter system are listed below:

Parameter	Value
Input Supply	Three-Phase AC Supply
DC Bus Voltage	Approximately 400 V DC
Output Voltage	230V RMS
Output Frequency	50 Hz
Inverter Type	Full-Bridge Single-Phase Inverter
Switching Technique	SPWM
Switching Frequency	20 kHz
Controller	STM32G4 Microcontroller
Output Filter	LC Low-Pass Filter
Switching Devices	MOSFETs
Protection Features	Overcurrent, Overvoltage, Fault Handling
Output Waveform	Pure Sine Wave

Table No 3.7.1 Key Design Parameters

3.8 Design Methodology Overview

The design methodology of the proposed inverter system follows a systematic approach involving circuit design, controller implementation, simulation, hardware development, testing, and performance evaluation [1], [4].

Initially, the overall system architecture and inverter topology were selected based on output requirements and power conversion efficiency. The rectifier stage and DC link section were designed to generate a stable high-voltage DC bus. The inverter stage was then designed using a full-bridge MOSFET configuration for high-frequency switching operation [1], [2].

The STM32G4 microcontroller was programmed to generate SPWM signals for inverter control. Voltage and current sensing circuits were integrated for monitoring and protection purposes. The LC filter parameters were selected to minimize harmonic distortion and improve waveform quality [3], [5], [8].

After hardware implementation, the system performance was tested under different operating conditions and load variations. Output voltage, waveform quality, switching performance, and inverter stability were analyzed to verify proper system operation [4], [6].

The proposed methodology provides a modular and scalable inverter design suitable for future improvements and practical power electronic applications [2], [4].

CHAPTER 4

PSIM SIMULATION

4.1 Introduction to Simulation

To validate the proposed system design before hardware implementation, simulations were carried out using PSIM software. Simulation helps in analyzing system behavior, verifying circuit operation, and evaluating output waveform quality under ideal conditions.

The simulation was performed in two stages. First, a single-phase inverter was simulated to understand SPWM-based DC-to-AC conversion. Then, the complete system including the three-phase rectifier, DC link, inverter, and LC filter was simulated to analyze overall performance.

4.2 SPWM Controlled Inverter Simulation

4.2.1 Circuit Description

The simulated circuit represents a single-phase full-bridge inverter using MOSFET switches. A DC input of approximately 400V is applied to the inverter.

SPWM is used as the switching technique, where a sinusoidal reference signal (50 Hz) is compared with a high-frequency triangular carrier signal (20 kHz). The resulting PWM signals control the switching of the MOSFETs in the H-bridge configuration.

The MOSFETs operate in diagonal pairs to produce alternating polarity across the load, resulting in a PWM-based AC waveform.

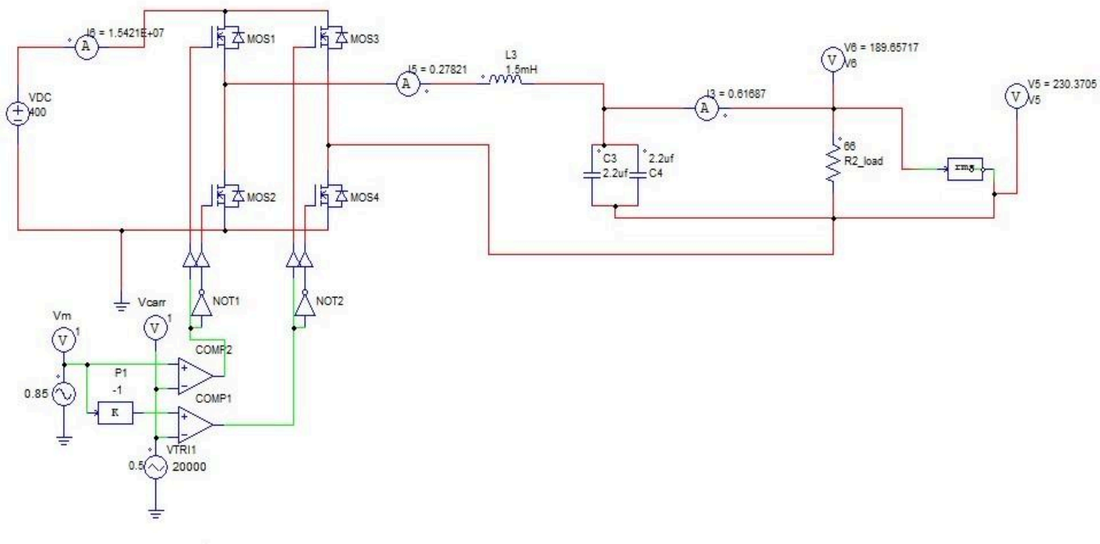


Fig 4.2.1 Inverter Simulation Circuit

4.2.2 Output Filtering

The output of the inverter consists of high-frequency PWM pulses, which contain harmonic components. To obtain a smooth sinusoidal waveform, an LC low-pass filter is used.

The inductor reduces rapid changes in current, while the capacitor smooths voltage variations. This combination effectively removes high-frequency components and retains the fundamental frequency.

4.3 Complete System Simulation (Rectifier + Inverter)

4.3.1 Rectifier Stage

The first stage of the simulation consists of a three-phase uncontrolled diode bridge rectifier. It converts the three-phase AC input into pulsating DC.

A capacitor bank is connected at the output of the rectifier to smooth the pulsating DC and form a DC link. This DC link provides a stable input voltage to the inverter stage.

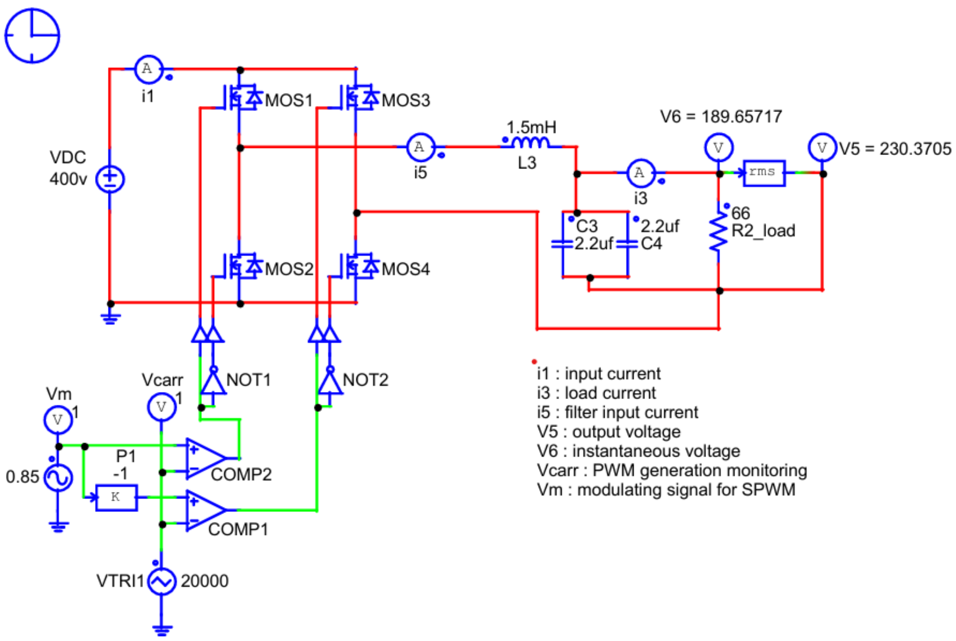


Fig 4.3.1 Rectifier Simulation Circuit

4.3.2 Inverter Stage

The DC link voltage is fed into the full-bridge inverter. The inverter uses SPWM signals to generate a high-frequency PWM waveform, which approximates a sinusoidal output.

The switching frequency is maintained at 20 kHz to ensure effective filtering and reduced harmonic distortion.

4.3.3 Output Filtering and Load

An LC filter is connected at the output of the inverter to remove high-frequency switching harmonics. The filtered output is applied to a resistive load.

The output waveform obtained after filtering closely resembles a sinusoidal waveform with reduced distortion.

4.4 Simulation Results and Analysis

4.4.1 Load Current

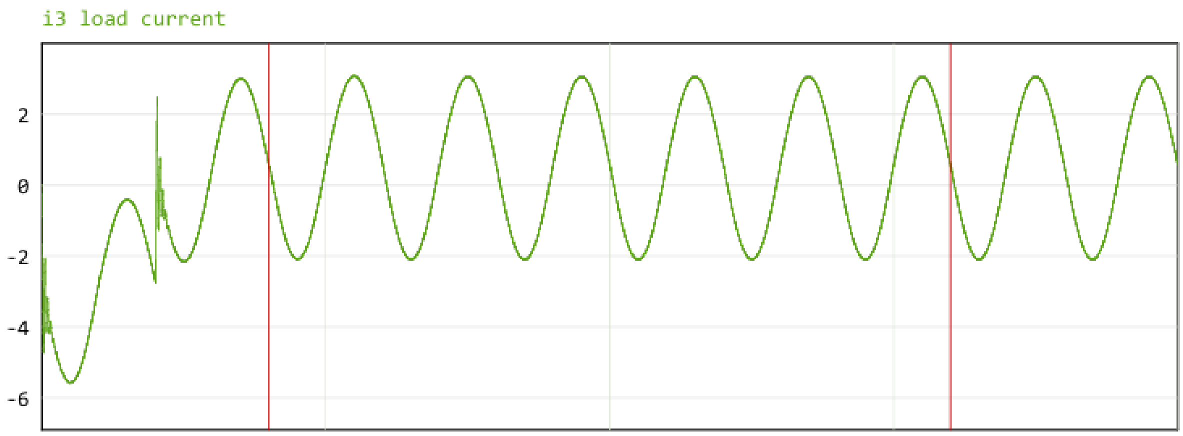


Fig 4.4.1 Load Current Waveform

The load current waveform shows a smooth sinusoidal pattern after filtering. This indicates that the LC filter effectively removes high-frequency components and allows only the fundamental frequency to pass.

4.4.2 Filter Input Current

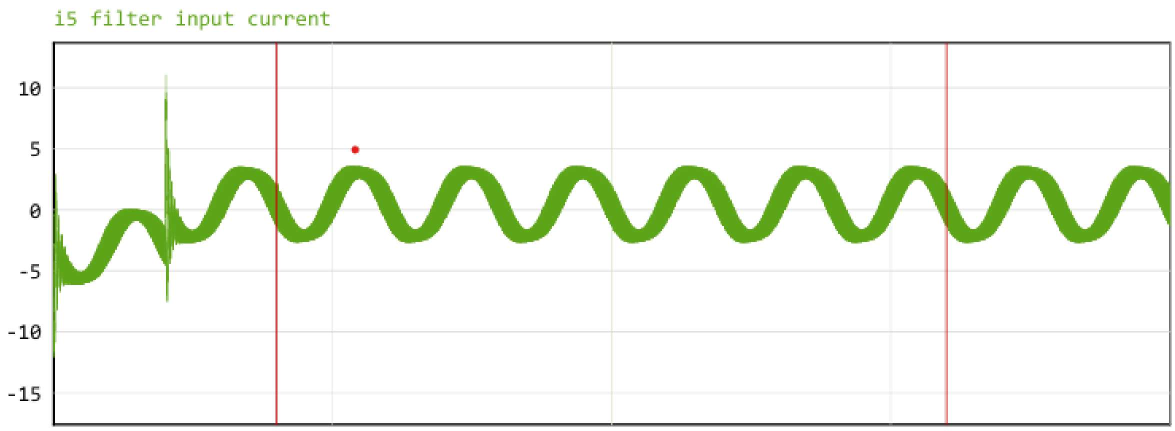


Fig 4.4.2 Filter Input Current Waveform

The current entering the filter contains high-frequency components due to switching action in the inverter. This waveform appears more distorted compared to the load current, confirming that the filter is actively removing unwanted harmonics.

4.4.3 Output Voltage

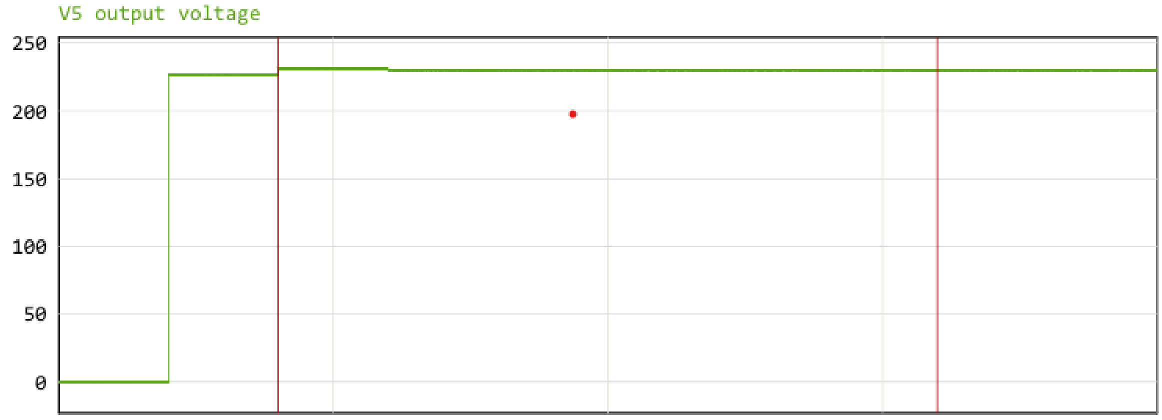


Fig 4.4.3 Output Voltage Waveform

The output voltage waveform after the LC filter is observed to be a stable sinusoidal waveform with an RMS value close to 230V. This confirms that the inverter is capable of producing the required output voltage.

4.4.4 Instantaneous Voltage

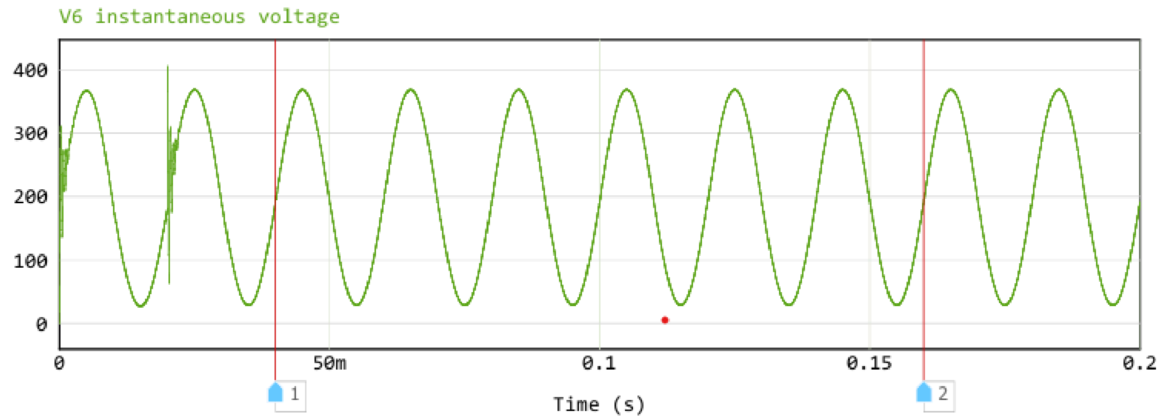


Fig 4.4.4 Instantaneous Voltage Waveform

The instantaneous voltage waveform shows the variation of voltage over time before filtering. It contains high-frequency switching components along with the fundamental frequency. After filtering, these high-frequency components are removed, resulting in a clean sinusoidal waveform.

4.5 Summary of Simulation

The simulation results confirm the proper operation of the two-stage inverter system. The rectifier successfully converts AC input into DC, and the inverter generates an AC output using SPWM control. The LC filter effectively reduces harmonic distortion and produces a clean sinusoidal waveform.

The simulation validates the design approach and provides confidence for proceeding to hardware implementation.

CHAPTER 5

HARDWARE IMPLEMENTATION

5.1 Introduction

After validating the system through simulation, the proposed two-stage inverter system was implemented in hardware. The implementation involves the design and integration of multiple subsystems, including the rectifier stage, DC link, inverter stage, gate driver circuits, auxiliary power supplies, sensing circuits, filtering stages, and control unit.

The hardware is designed using a modular approach, where power and control circuits are separated to improve safety, reduce noise interference, and simplify debugging. Proper consideration is given to isolation, thermal performance, and PCB layout to ensure reliable operation at high power levels.

5.2 Rectifier Circuit Design

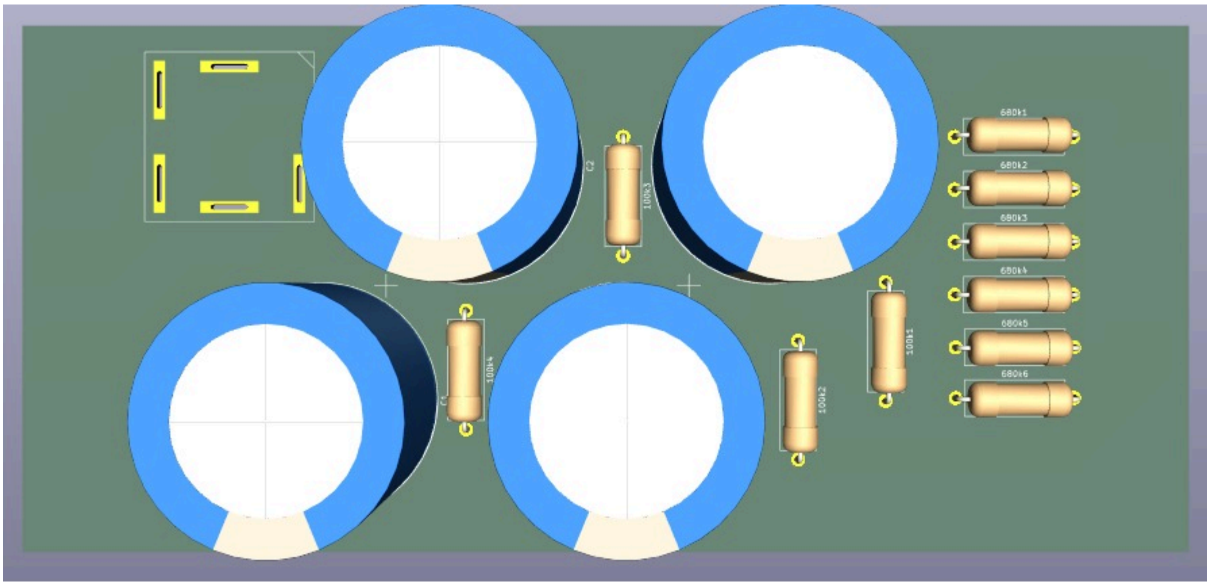


Fig 5.2.1 Rectifier PCB -1

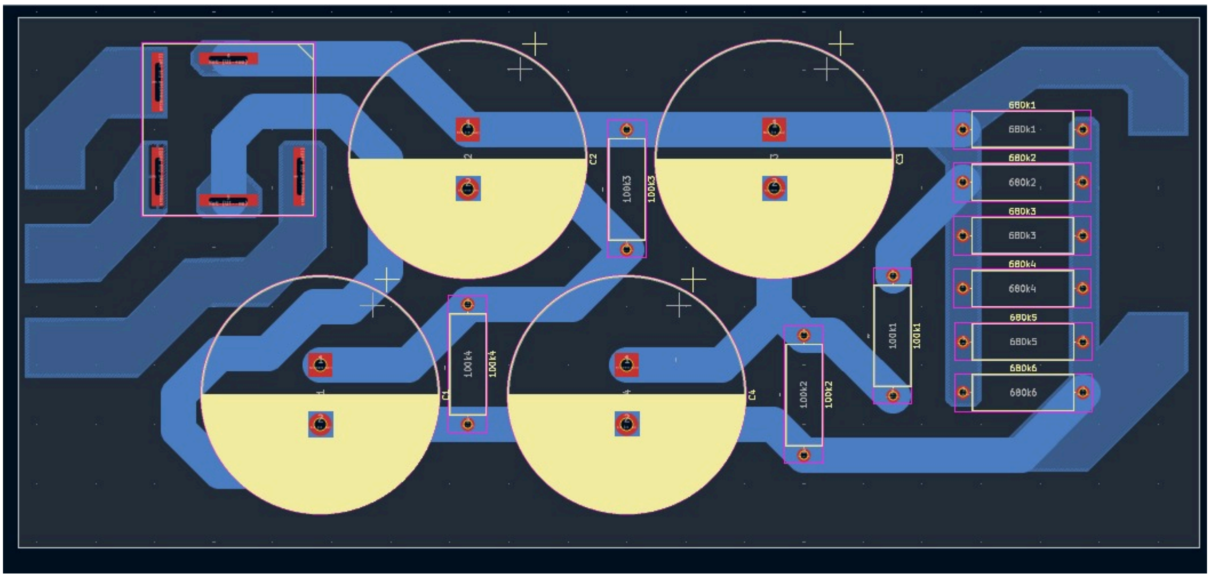


Fig 5.2.2 Rectifier PCB -2

Description:

The rectifier stage consists of a three-phase uncontrolled diode bridge, which converts the AC input supply into pulsating DC. The bridge is formed using six power diodes arranged in a full-wave configuration, where each phase contributes to the output during different intervals of the AC cycle.

Working Principle:

During operation, at any instant, two diodes conduct, one from the positive group and one from the negative group, allowing current to flow from the highest positive phase to the most negative phase. This results in a pulsating DC output with reduced ripple compared to single-phase rectification.

Key Components:

1. Power Diodes (D1–D6): Handle high voltage and current
2. Input Terminals: Three-phase AC supply
3. Output Terminals: DC output to DC link

Importance in System:

This stage provides the required DC voltage for the inverter. Its simplicity, reliability, and ruggedness make it suitable for high-power applications.

5.3 DC Link Design

Description:

The DC link connects the rectifier stage to the inverter and acts as an intermediate energy storage element. It consists of a capacitor bank designed to smooth the pulsating DC output from the rectifier.

Working Principle:

The capacitors charge during voltage peaks and discharge during voltage dips, thereby reducing ripple and maintaining a nearly constant DC voltage. This stabilized voltage is critical for proper inverter operation.

Design Considerations:

1. High voltage rating (≈400V)
2. High ripple current handling
3. Low Equivalent Series Resistance (ESR)
4. Proper spacing and PCB layout

Importance in System:

A stable DC link ensures consistent inverter performance, reduces stress on switching devices, and improves output waveform quality.

5.4 Inverter Circuit Design

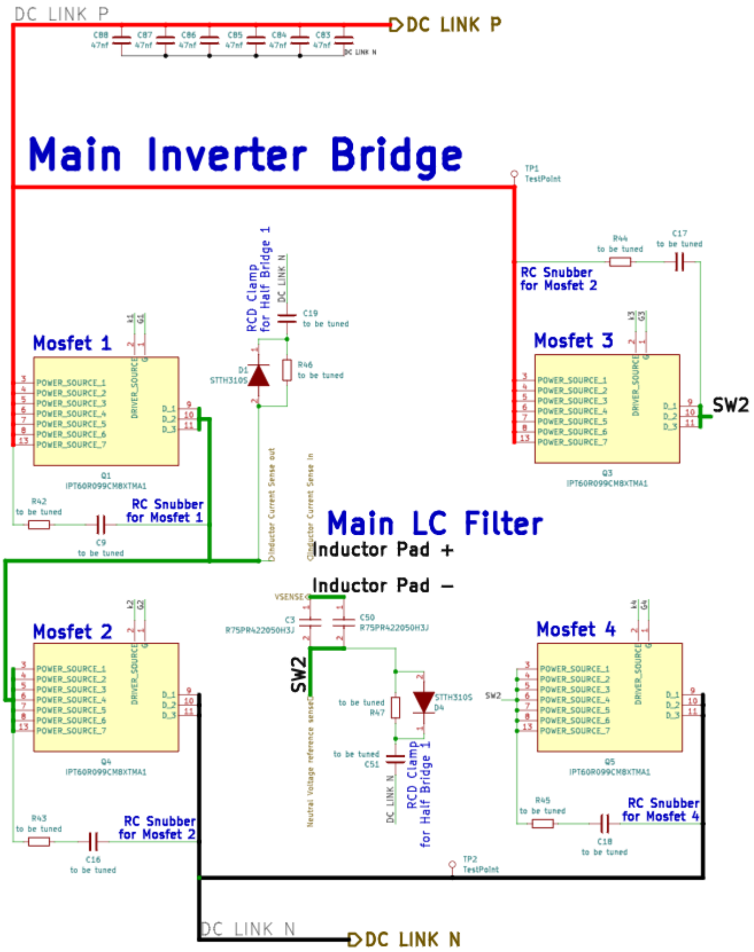


Fig 5.4.1 Inverter Schematic Diagram

Description:

The inverter stage converts DC into AC using a full-bridge configuration consisting of four N-channel MOSFETs. The MOSFETs are controlled using SPWM signals generated by the microcontroller.

Working Principle:

The MOSFETs operate in diagonal pairs (Q1–Q4 and Q2–Q3). By switching these pairs alternately, the polarity of the voltage across the load is reversed, producing an AC waveform. The switching is performed at a high frequency (20 kHz) to enable efficient filtering.

Protection Features:

1. RC Snubber Circuits: Reduce voltage spikes
2. RCD Clamp: Protect against overvoltage
3. Decoupling Capacitors: Stabilize DC input

Importance in System:

This is the core stage responsible for generating AC output from DC power.

5.5 Gate Driver Circuit

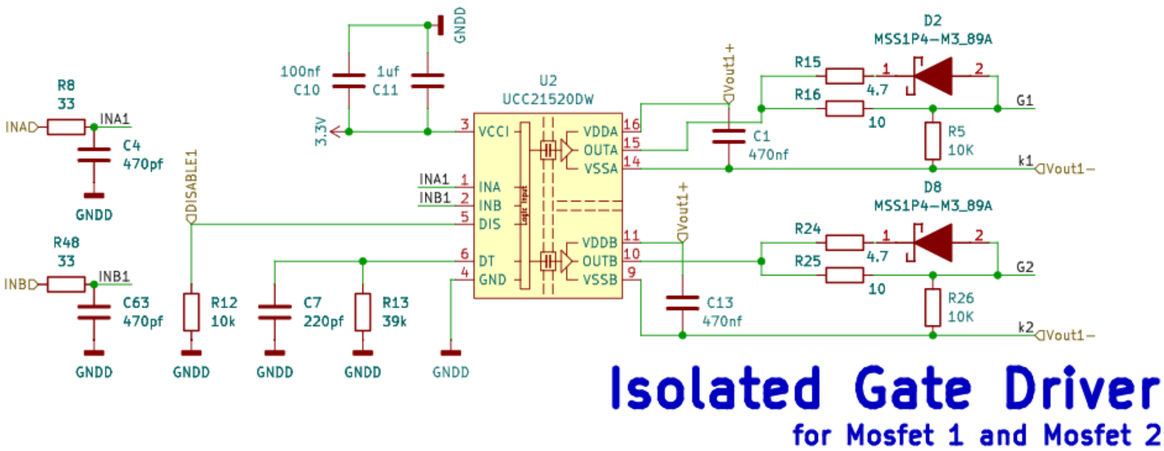


Fig 5.5.1 Isolated Gate Driver Schematic Diagram -1

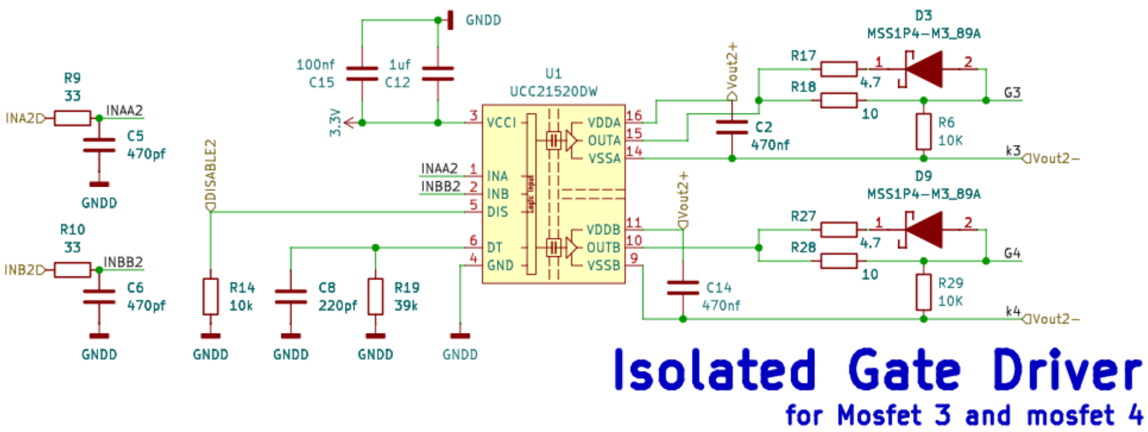


Fig 5.5.2 Isolated Gate Driver Schematic Diagram -2

Description:

The gate driver circuit is used to drive the MOSFETs with appropriate voltage and timing. The UCC21520 isolated dual-channel gate driver is used for this purpose.

Working Principle:

The driver receives PWM signals from the microcontroller and amplifies them to drive the MOSFET gates. It also provides electrical isolation between the control and power stages.

Key Features:

1. Reinforced isolation
2. Dead-time control to prevent shoot-through
3. High switching speed
4. Separate isolated supplies

Importance in System:

Ensures safe, efficient, and synchronized switching of MOSFETs.

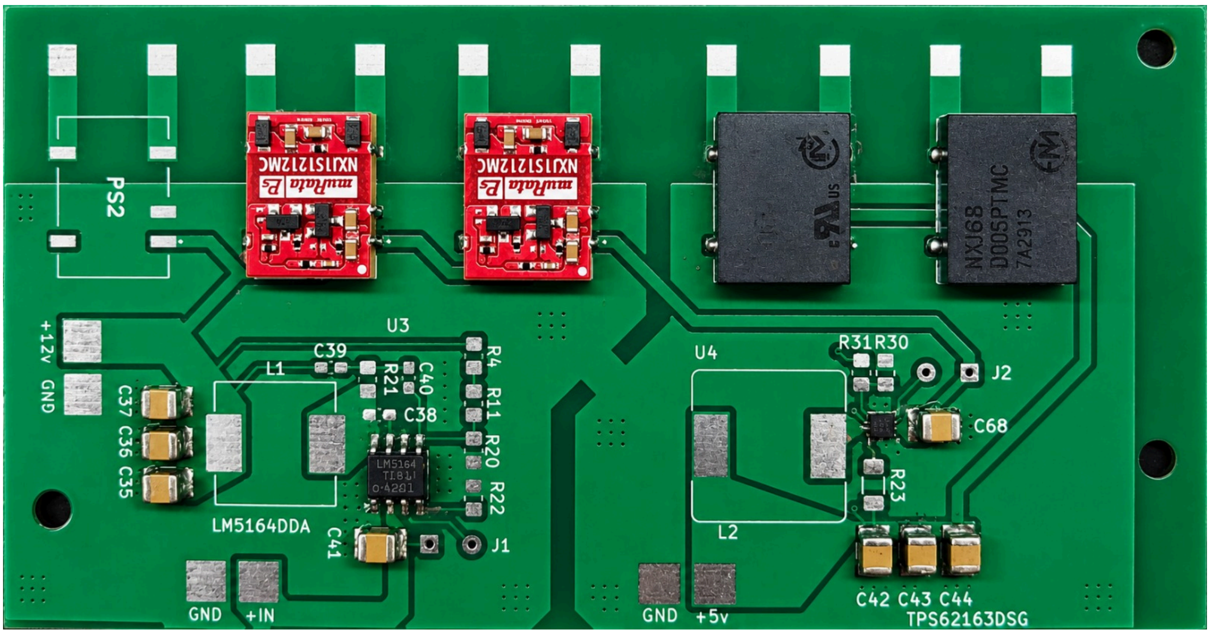


Fig 5.5.3 Isolated Gate Driver PCB

5.6 Auxiliary Power Supply Design

5.6.1 72V/High Voltage to 12V Converter

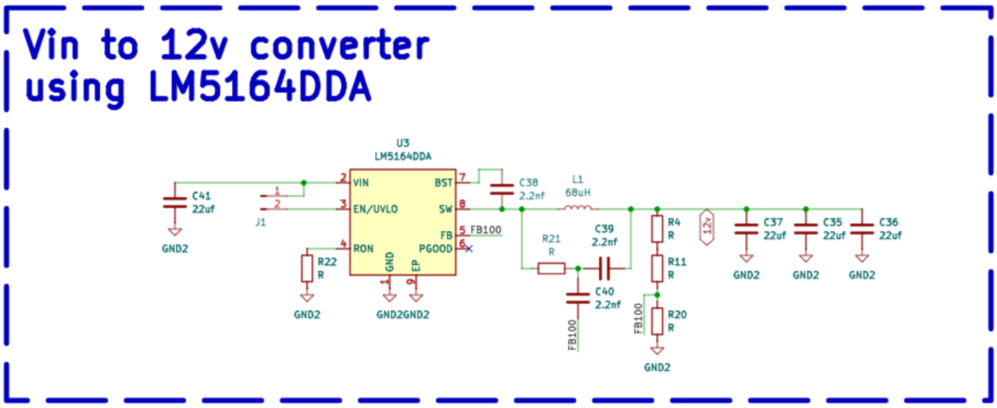


Fig 5.6.1 72V/High Voltage to 12V Converter Schematic Diagram

This converter steps down the high input voltage to 12V, which is used to power gate drivers and intermediate circuits.

5.6.2 12V to 5V Converter

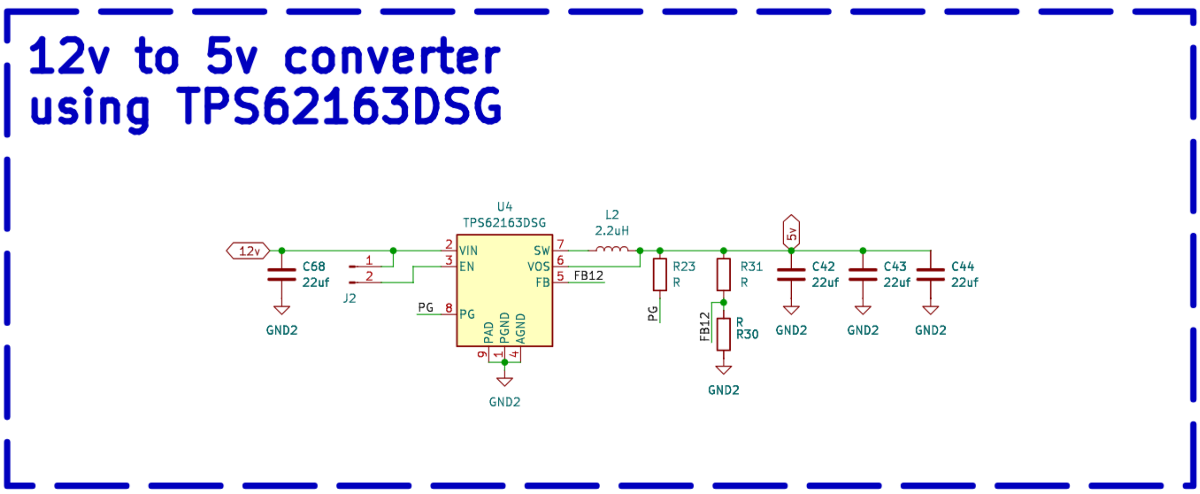


Fig 5.6.2 12V to 5V Converter Schematic Diagram

The 12V to 5V buck converter provides a stable 5V supply for low-voltage components such as sensors and auxiliary circuits.

5.6.3 Isolated Power Supply

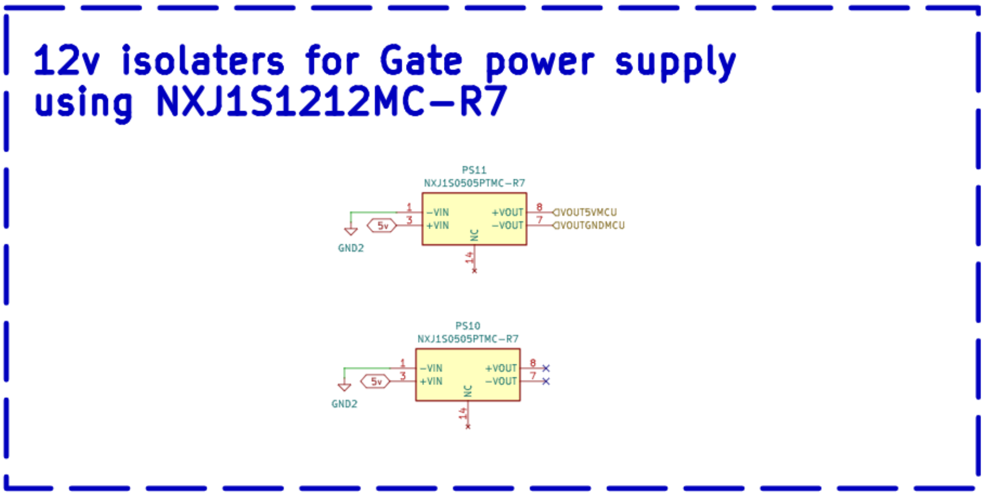


Fig 5.6.3 Isolated Power Supply Schematic Diagram -1

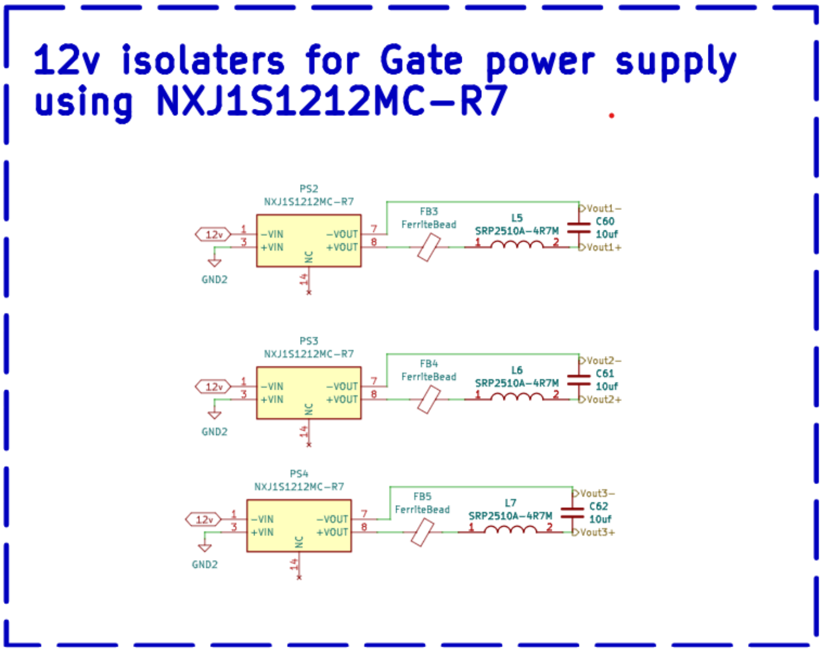


Fig 5.6.4 Isolated Power Supply Schematic Diagram -2

Isolated DC-DC converters are used to provide an isolated ±12V supply for gate drivers. This ensures safe operation and prevents noise coupling between control and power circuits.

5.7 Sensing and Measurement Circuits

5.7.1 Output Current Measurement

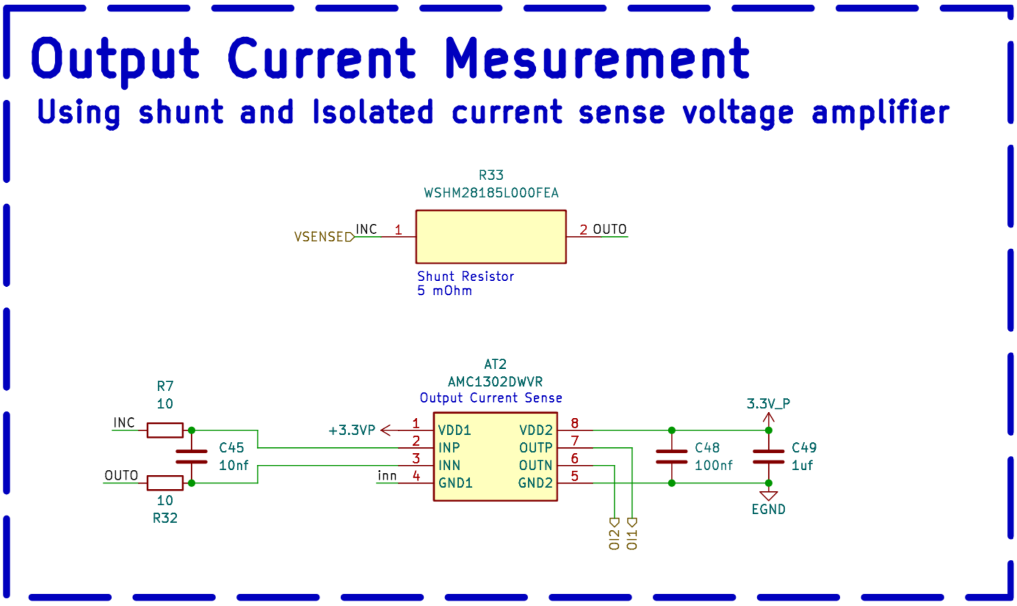


Fig 5.7.1 Output Current Measurement Schematic Diagram

A shunt resistor combined with an isolated amplifier is used to measure output current. The amplifier provides accurate measurement while maintaining electrical isolation.

5.7.2 Input Current Measurement

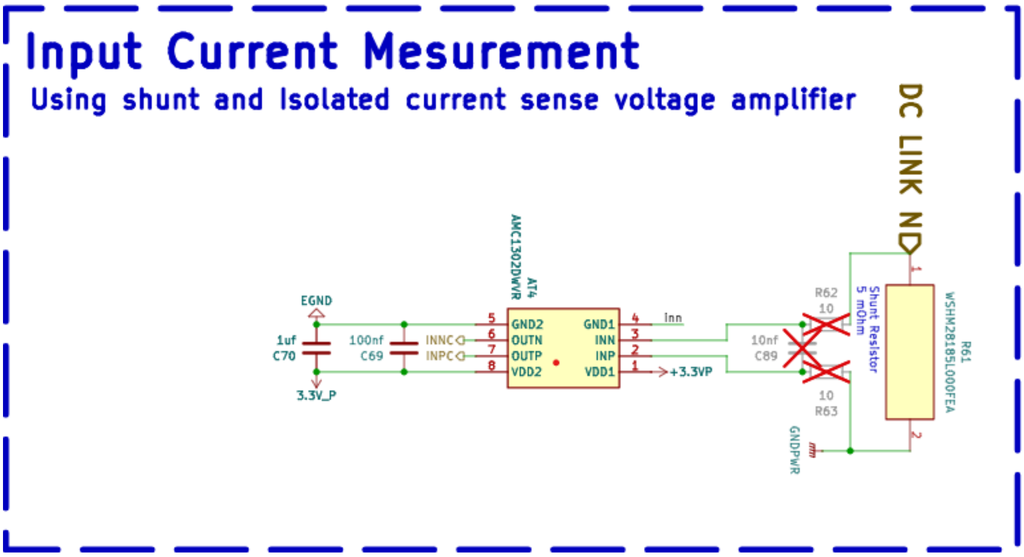


Fig 5.7.2 Input Current Measurement Schematic Diagram

Similar to output sensing, the input current is measured using a shunt resistor and isolated amplifier for monitoring and protection.

5.7.3 Voltage Measurement

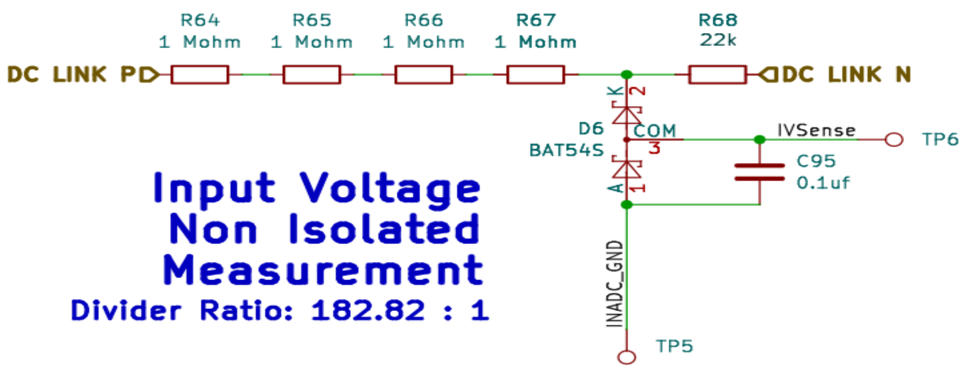


Fig 5.7.3 Input Voltage Measurement Schematic Diagram

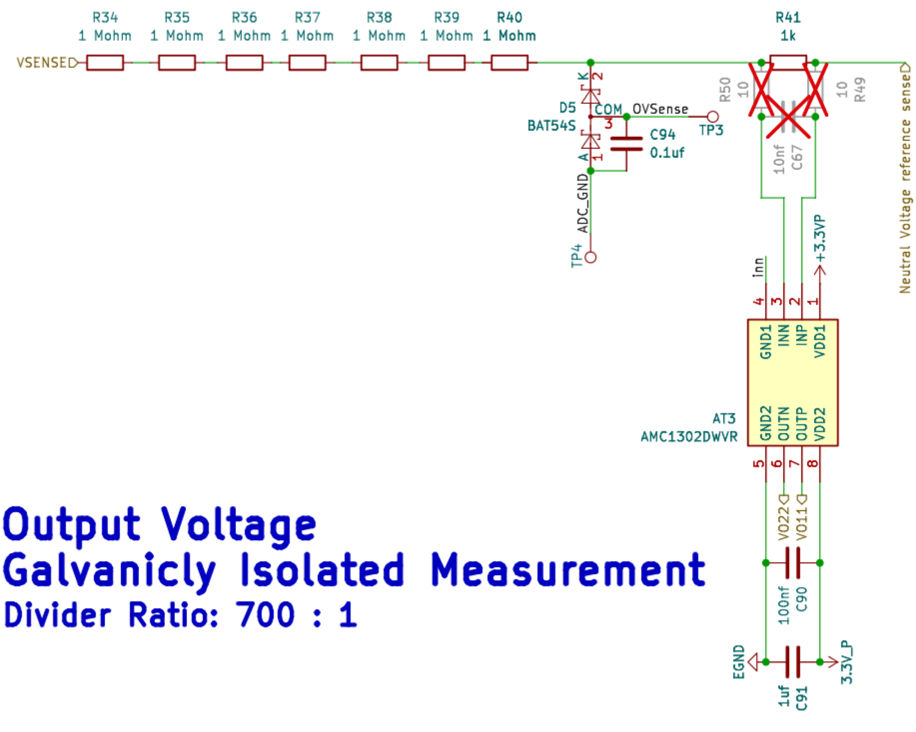


Fig 5.7.4 Output Voltage Measurement Schematic Diagram

Voltage measurement circuits are used to scale down high voltages to safe levels for the microcontroller. Isolation is used where necessary to ensure safety.

5.8 Output Filter and EMI Filter

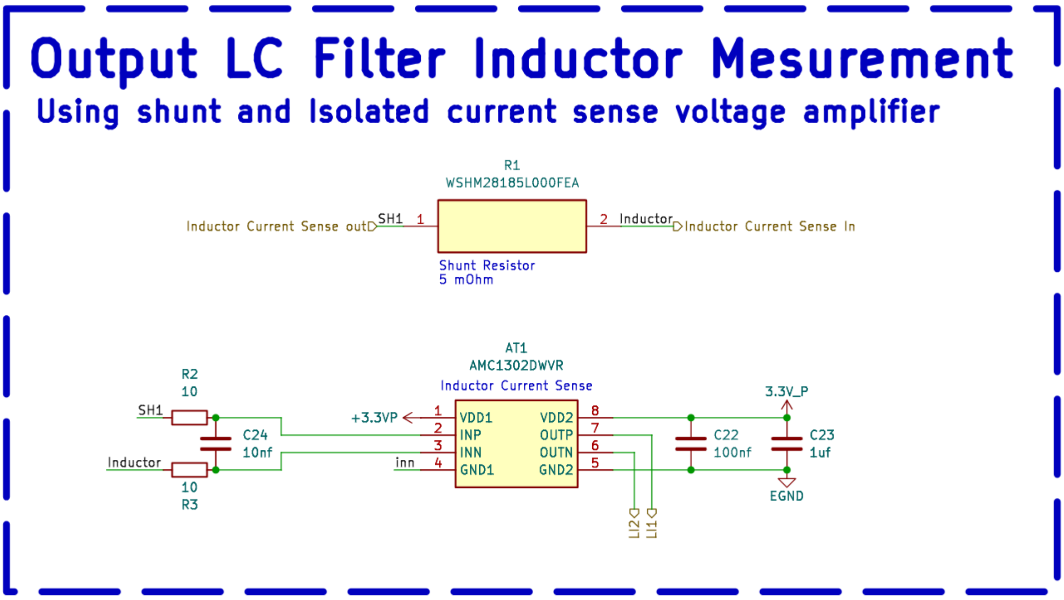


Fig 5.8.1 Output LC Filter Schematic Diagram

The LC filter is used to convert the PWM waveform into a smooth sinusoidal output by removing high-frequency components.

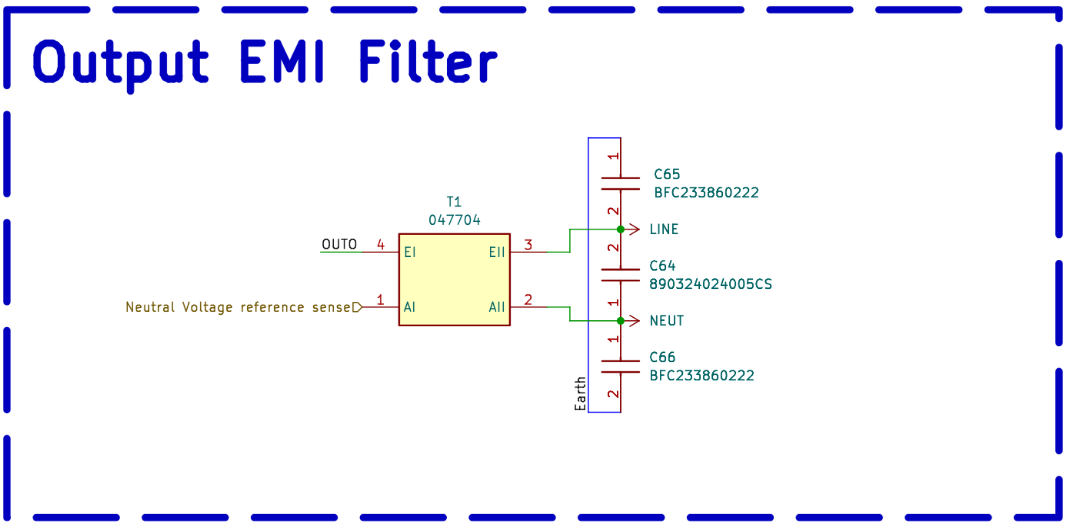


Fig 5.8.2 EMI Filter Schematic Diagram

An EMI filter is also used to suppress electromagnetic interference and ensure compliance with noise standards.

5.9 Microcontroller and Control Interface

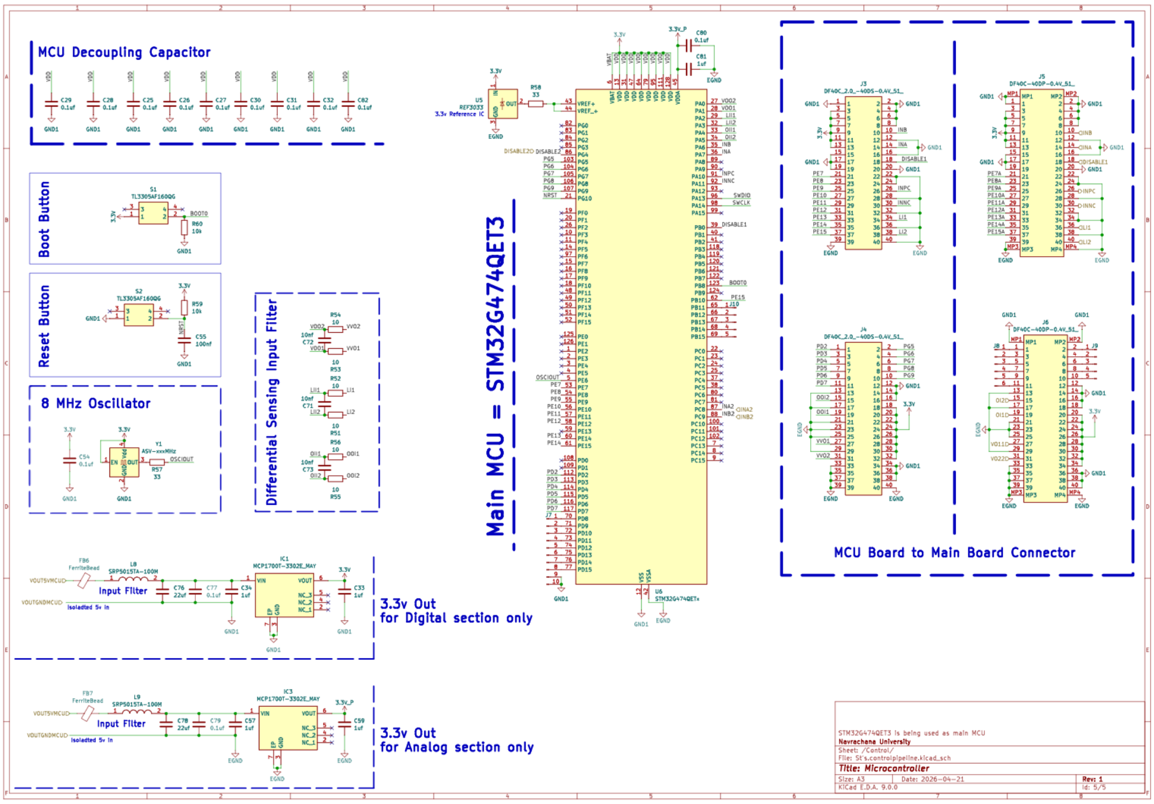


Fig 5.9.1 Microcontroller and Control Interface Schematic Diagram

The above schematic represents the microcontroller and control interface of the inverter system. The STM32G474QET3 microcontroller acts as the central control unit, responsible for generating SPWM signals, processing feedback signals, and controlling the switching of the inverter.

The circuit includes supporting components such as decoupling capacitors, clock oscillator, reset circuitry, input filtering, voltage regulation, and connector interfaces to ensure stable and reliable operation.

5.9.1 STM32G4 Microcontroller

The STM32G4 microcontroller is the main processing unit of the system. It generates high-frequency PWM signals based on SPWM logic and controls the inverter switching through gate drivers.
It also receives feedback signals such as output voltage and current, enabling closed-loop control and system protection.

5.9.2 Decoupling Capacitor Network

Multiple decoupling capacitors are connected across the power supply pins of the microcontroller. These capacitors filter high-frequency noise and stabilize the supply voltage.

Smaller capacitors handle high-frequency noise, while larger capacitors provide bulk energy storage. This ensures reliable operation and prevents unwanted fluctuations during switching.

5.9.3 Clock Circuit (8 MHz Oscillator)

An external 8 MHz crystal oscillator is used to provide an accurate clock signal to the microcontroller.

This clock is essential for precise timing of PWM signals and overall system synchronization. Stable clock operation directly affects switching accuracy and output waveform quality.

5.9.4 Reset Circuit

The reset circuit ensures that the microcontroller starts in a known state during power-up or fault conditions.

A push-button reset allows manual restarting of the system, while the resistor-capacitor network ensures proper reset timing.

5.9.5 Boot Configuration Circuit

The boot configuration circuit determines the startup mode of the microcontroller. It ensures that the program is executed from internal memory during normal operation.

5.9.6 Input Signal Conditioning (Filtering Section)

Input signals from sensing circuits pass through filtering networks before reaching the microcontroller.

These filters remove noise and ensure that only clean signals are processed. This improves measurement accuracy and prevents incorrect control decisions due to noise.

5.9.7 Voltage Regulation (3.3V Supply)

The microcontroller operates at 3.3V, which is derived from regulated power supply circuits.

Separate regulation is provided for digital and analog sections to reduce noise interference. This improves ADC accuracy and overall system performance.

5.9.8 MCU to Main Board Connector Interface

The connector interface links the microcontroller board with the main power board.

It carries:

1. PWM signals → gate drivers
2. Feedback signals → MCU
3. Power supply lines

This modular design allows easy debugging, replacement, and system scalability.

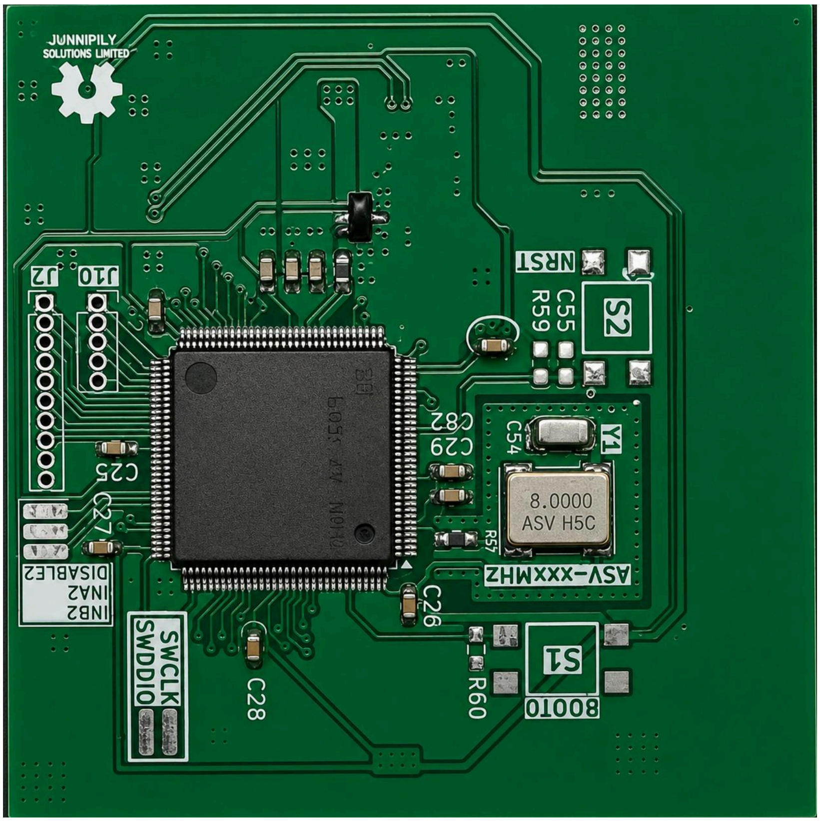


Fig 5.9.2 MCU PCB

5.10 PCB Design and Implementation

The system is implemented using a modular PCB design to separate power and control circuits. Proper grounding, isolation, and routing techniques are used to minimize noise and improve performance.

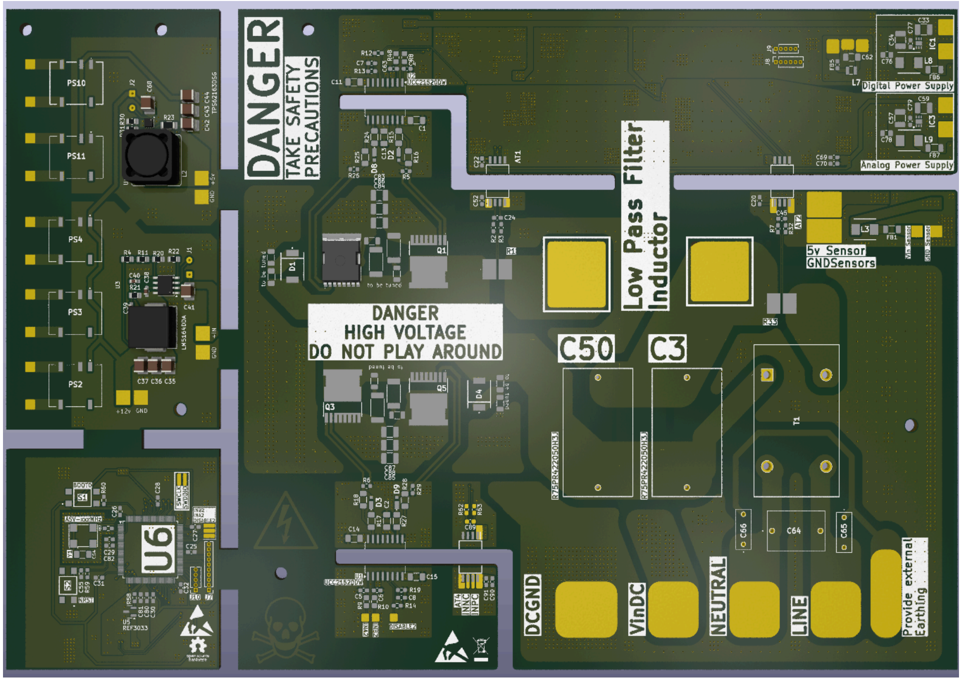


Fig 5.10.1 Inverter PCB -1

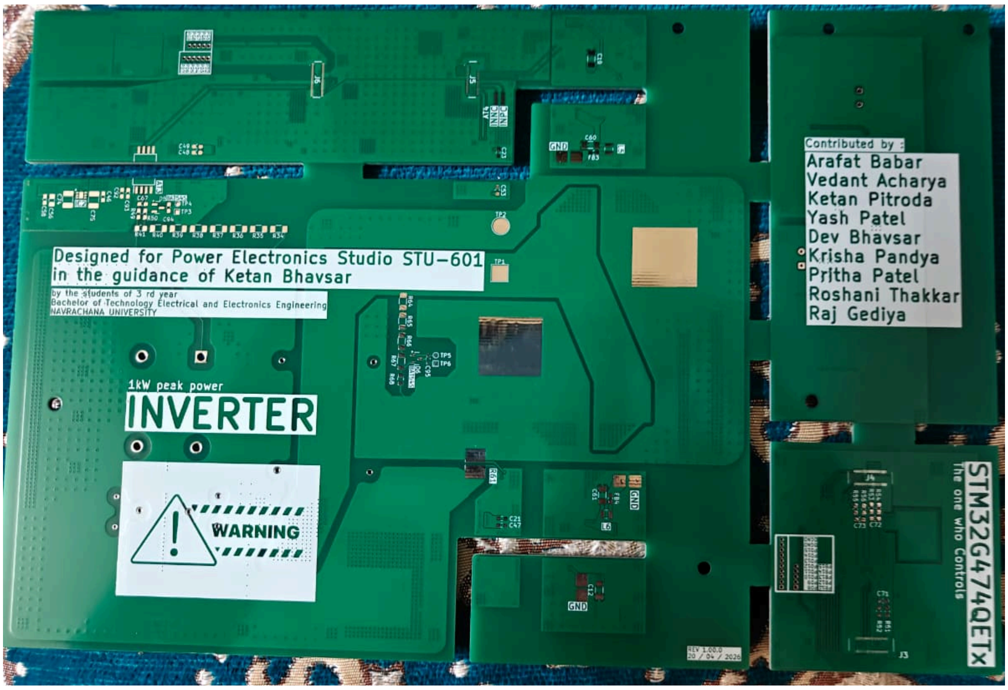


Fig 5.10.2 Inverter PCB -2

5.11 Summary of Components

Component	Part Number	Rating	Purpose	Used In
Power MOSFET	IPT60R099CM8XTMA1	600V, 30A	Main switching device for the inverter, handles high voltage DC bus	Full-Bridge Inverter
Power MOSFET	IPT034N15NM6ATMA1	150V, 194A	High current switching is used in lower voltage stages or auxiliary circuits	Switching Stage
Current Sense Resistor	WSHM28185L000FEB	5 mΩ, 7W, 1%	Measures current for monitoring and protection	Protection Circuit
Oscillator	ASV-8.000MHZ-EJ-T	8 MHz	Provides a clock signal for the microcontroller	Control Circuit
DC-DC Converter (Isolated)	NXJ1S0505PTMC-R7	5V, 0.2A	Provides an isolated power supply for the control/gate driver	Control & Driver Circuit
Buck Converter	TPS62163DSGT	High frequency (2.25 MHz), step-down regulator	Converts higher DC voltage to lower required levels	Power Supply Stage
LDO Regulator	MCP1700-3302E	3.3V, 250mA	Provides a stable voltage for the microcontroller	Control Circuit
Microcontroller	STM32G4	High-speed MCU with PWM support	Generates SPWM signals and controls inverter operation	Control Unit
Rectifier	Diode Bridge (3-phase)	≥400V, suitable current rating	Converts AC input to DC	Rectifier Stage

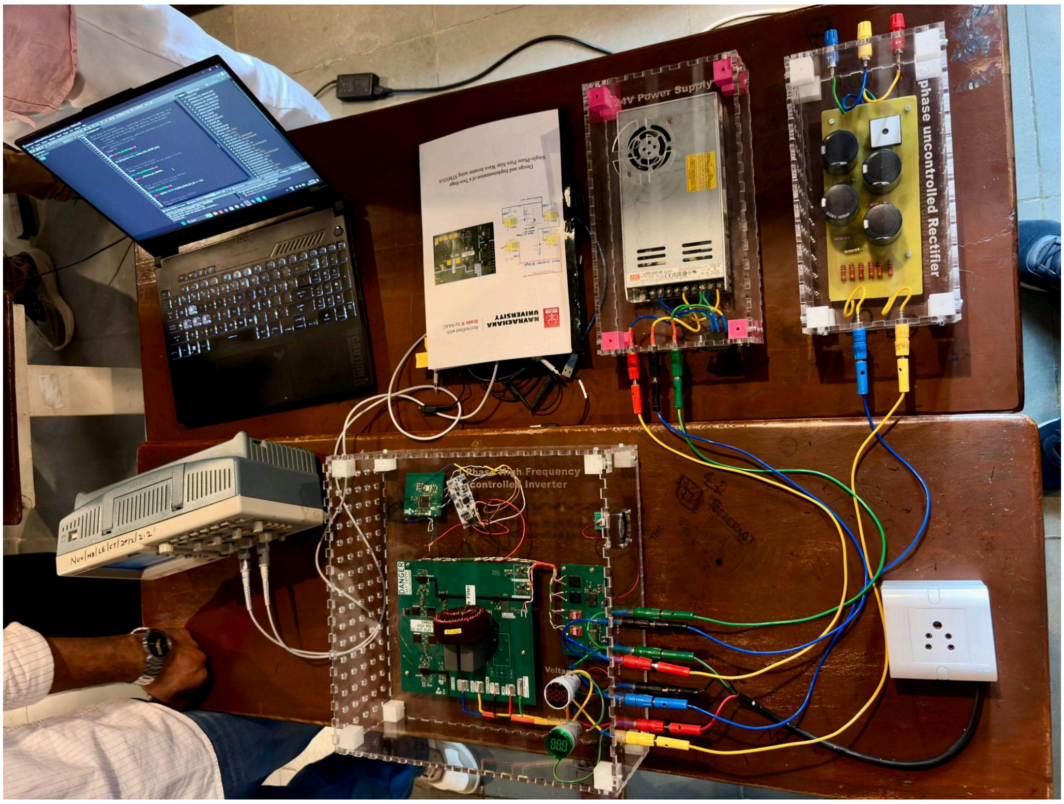
DC Link Capacitor	Electrolytic Capacitor	~400V rating, high capacitance	Smoothens DC output and reduces ripple	DC Link
Gate Driver	Gate Driver IC (isolated/non-isolated)	Suitable gate voltage (10–15V)	Drives MOSFET gates for switching	Driver Circuit
Output Filter	LC Filter	Designed for 50 Hz output	Removes harmonics and smoothens waveform	Output Stage
Load	Resistive Load	As per testing (e.g., 1kW)	Testing and performance evaluation	Output

Table No 5.11.1 Summary of Components

CHAPTER 6
RESULT AND ANALYSIS

Experimental Test Bench Description:

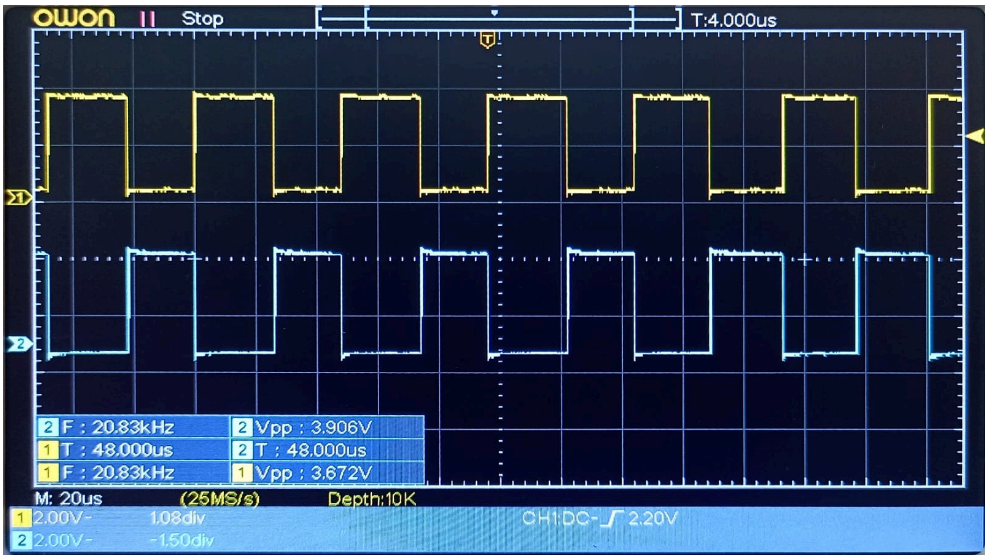
1. Three-Phase Uncontrolled Rectifier Module: Converts the Single-phase AC input into a pulsating DC output. The module is housed in a transparent acrylic enclosure and feeds the DC bus voltage to the inverter stage.
2. High-Frequency Variable Inverter PCB: The primary hardware prototype implementing the full-bridge MOSFET inverter topology with SPWM control. The board integrates the gate driver circuits (UCC21520), LC output filter, snubber protection, and auxiliary power supply sub-circuits.
3. Auxiliary (AV) Power Supply Module: Provides regulated low-voltage DC supply for the control circuitry, gate drivers, and sensing subsystems.
4. STM32G4 Microcontroller Control Board: Generates 20 kHz SPWM switching signals and manages voltage/current sensing, PI control, dead-time insertion, and fault protection.
5. Tektronix TBS1022 DSO (Scientech): Two-channel digital storage oscilloscope used for real-time waveform acquisition and measurement of switching and output signals.
6. Laptop Workstation: Runs the embedded programming and real-time monitoring interface for the STM32G4 controller via USB/SWD programming port.



Two-Stage Single-Phase Pure Sine Wave Inverter System

Oscilloscope Configuration and Measurement Setup

Parameter	Channel 1 (CH1)	Channel 2 (CH2)
Signal Observed	SPWM Gate Drive Signal	Output Voltage Response
Coupling Mode	DC	DC
Vertical Scale (V/div)	1.00 V	100 mV
Bandwidth Limit	OFF (Full 25 MHz)	OFF (Full 25 MHz)
Probe Attenuation	1X Voltage	1X Voltage
Invert Function	OFF	OFF
Time Base (ms/div)	5.00 ms	5.00 ms
Total Acquisition Window	50 ms (10 divisions)	50 ms (10 divisions)
Trigger Position (M Pos)	3.600 ms	3.600 ms
CH2 DC Level Reading	—	−33.9 mV
Acquisition Status	Stop (Frozen waveform)	Stop (Frozen waveform)
Date / Time Stamp	22-May-2026 14:07	22-May-2026 14:07



Tektronix TBS1022 DSO connected to inverter PCB output terminals

The time base of 5.00 ms/div provides a total acquisition window of 50 ms across the ten horizontal divisions of the display. This window spans exactly one full period of the 50 Hz fundamental output frequency, allowing simultaneous observation of both the high-frequency switching waveform (20 kHz SPWM) and the complete 50 Hz output cycle. DC coupling was employed on both channels to ensure that any DC offset components present in the circuit are accurately represented without signal suppression.

APPENDIX

Appendix A – Block Diagram of Proposed System

This appendix contains the overall block diagram of the proposed Two-Stage Single-Phase Pure Sine Wave Inverter using STM32G4. The block diagram illustrates the complete power flow and control architecture of the system including:

1. Three-phase AC input supply
2. Three-phase rectifier stage
3. DC link capacitor bank
4. Full-bridge inverter stage
5. STM32G4 control board
6. SPWM generation
7. LC output filter
8. Voltage and current sensing circuits

The block diagram is used to understand the interconnection between various stages of the inverter system and the overall system operation.

Appendix B – PSIM Simulation Circuits

This appendix includes the PSIM simulation models developed for validating the inverter design before hardware implementation. The following simulation circuits are included:

1. SPWM controlled full-bridge inverter
2. Three-phase rectifier circuit
3. Complete two-stage inverter simulation
4. LC filter implementation

The simulations were used to analyze switching performance, waveform quality, harmonic reduction, and inverter output characteristics.

Appendix C – Output Waveforms

This appendix contains the waveform results obtained during simulation and testing. The included waveforms are:

1. Load current waveform
2. Filter input current waveform
3. Output voltage waveform
4. Instantaneous voltage waveform
5. SPWM switching pulses

The waveform analysis confirms the successful generation of a pure sine wave output with reduced harmonic distortion.

Appendix D – PCB Layouts and Hardware

This appendix contains the PCB layouts and hardware design files used in the implementation of the inverter system. The following hardware sections are included:

1. Rectifier PCB
2. Inverter PCB
3. Gate driver PCB
4. STM32G4 control board PCB
5. Auxiliary power supply PCB

The appendix also includes hardware photographs and assembled system images for reference.

Appendix E – Schematic Diagrams

This appendix contains detailed schematic diagrams of all major subsystems used in the project, including:

1. Full-bridge inverter schematic
2. Gate driver circuit schematic
3. Voltage sensing circuits
4. Current sensing circuits
5. DC-DC converter circuits

6. LC filter schematic
7. EMI filter schematic
8. STM32G4 control interface schematic

These schematics provide complete circuit-level understanding of the implemented inverter system.

Appendix F – Component Datasheets

This appendix contains datasheets of important electronic components used in the project for design reference and specification verification. The major datasheets included are:

1. STM32G4 Microcontroller
2. UCC21520 Gate Driver IC
3. Power MOSFETs
4. DC-DC Converter Modules
5. Current Sense Amplifiers
6. Voltage Regulators
7. Electrolytic Capacitors

The datasheets were referred to during circuit design, component selection, and hardware implementation.

Appendix G – Source Code and PWM Configuration

This appendix contains the embedded C source code developed for the STM32G4 microcontroller. The source code includes:

1. SPWM generation logic
2. Timer configuration
3. ADC configuration
4. Voltage and current sensing routines
5. Protection algorithms
6. PWM dead-time insertion
7. Interrupt handling routines

The appendix also contains details of PWM frequency selection, modulation index configuration, and control logic implementation.

Appendix H – Testing and Safety Precautions

This appendix describes the testing procedures and safety precautions followed during hardware implementation and experimentation. The following precautions were considered:

1. Proper electrical isolation
2. Safe grounding techniques
3. High-voltage handling precautions
4. Use of insulated probes and equipment
5. Controlled startup using variac
6. Thermal monitoring of switching devices

These precautions ensured safe operation during inverter testing and performance evaluation.

Appendix I - Groups of members

Group A: Project Team, Power Electronics & Drives Studio (STU 601)

“The team successfully designed and implemented a modular two-stage pure sine wave inverter system capable of generating a stable 230 V RMS, 50 Hz single-phase AC output. The project encompassed a three-phase full-bridge rectifier stage, a 400 V DC bus design, a full-bridge MOSFET inverter controlled via 20 kHz SPWM signals, LC output filtering, isolated gate driver circuits, and PI-based closed-loop control - all implemented on custom-designed PCBs and verified through PSIM simulation and hardware testing using a Tektronix TBS1022 DSO.”

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